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Avere Systems Special Edition

Cloud NAS

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Learn to:

- Optimize your NAS environments
- Expand the abilities of your storage infrastructure
- Connect seamlessly to cloud storage

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A V **E** R E

Alex Nikitin



About Avere Systems

Avere Systems is radically changing the economics of data storage.

Avere solutions give companies the ability to put an end to the rising cost and complexity of data storage by allowing customers the freedom to store files anywhere in the cloud or on premise without sacrificing the performance, availability, or security of their data.

Enterprises understand the potential game-changing economics provided by cloud storage, but limitations such as an unfamiliar object interface, unreliable performance due to high latency, and concerns regarding availability of data have prevented many from taking advantage of the tremendous cost savings that the cloud offers. Avere Cloud NAS helps enterprises overcome these hurdles by providing a purpose-built enterprise solution that integrates their existing storage systems with the cloud without sacrificing the performance, availability, or security of their data.

In a broad range of industries from media and entertainment to financial services, from software development to oil and gas research, from life sciences to the Web, customers are using Avere solutions to bridge the gap between cloud economics and performance. Doing so results in productivity gains that translate to higher revenues and happier customers.

By delivering amazing performance at a fraction of the cost and ecological footprint of traditional storage, Avere has reinvented storage.

Cloud NAS

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by Alex Nikitin

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Introduction

The World Wide Web's influence on traditional business computing has caused an explosive rate of growth in the data that businesses use to make money. The ability to capture data effectively, store it, access it, and use it quicker and more efficiently than other companies is now the challenge of every corporation, large and small. Leveraging the advantages of “cloud” without having to throw out your existing computing and storage infrastructures is the next major shift in traditional in-house data-center-centric storage practices. If increasing data-management worries and confusion over how to take advantage of cloud storage today are things that keep you up at night, this book is for you.

About This Book

Cloud NAS For Dummies, Avere Systems Special Edition, gets you up to speed on Network Attached Storage (NAS) and Cloud NAS. Discover how cloud storage technologies, combined with Avere Cloud NAS, converge and optimize your adaptation of cloud technologies with your existing NAS storage infrastructure.

Icons Used in This Book

Throughout this book, special icons call attention to important information. You'll definitely want to take note!



This icon points out information that may well be worth committing to memory, at least that's the intention!



Here, I call attention to some quick technical stuff. You can skip it if you want.



This icon points out helpful suggestions and useful nuggets of information.



Danger! This helpful alert offers practical advice to help you avoid making potentially costly mistakes.

Beyond the Book

Avere Cloud NAS is reinventing the way you store and access critical business data. Check out www.storagereinvented.com or www.averesystems.com for further information on Cloud NAS.

Where to Go from Here

Hey, it's your book so you can start anywhere you like, but you might want to take a look at Chapters 1 and 2 first to get the backdrop for NAS and the cloud. Otherwise, feel free to look for a topic you're particularly interested in or just dive in anywhere the book leads you. It's entirely up to you.

Chapter 1

Understanding NAS

In This Chapter

- ▶ Reviewing Network Attached Storage
- ▶ Managing data growth
- ▶ Looking for ways to resolve overloaded NAS systems

Network Attached Storage (NAS) has been around for quite some time. In fact, as soon as early system architects decided to build a network between their computers so that they could share information, they essentially built the precursor to today's NAS systems. Once these architects sent some bits and bytes from one computer to another, they decided it would be a great idea to keep that data around for a while — which led to the remote storage of data, separate from the computer that actually created it. It also led to the accumulation of data that didn't need to be stored for long, if at all. Because people like to play it safe and keep data around just in case, the problem of data management was born.

In this chapter, I review what NAS is, give you some insight into how managing data nowadays can be difficult, and offer some ideas and guidelines for managing vast amounts of data.

Getting to the Basics of NAS Storage

NAS is a method of presenting a shared area for many different computers or users to store their files via a network. Creating a NAS system doesn't take much more than these components:

- ✓ A network, which is the key to being able to access data from just about anywhere in a company as long as a communication link exists between the NAS server and your clients' computers.
- ✓ The physical storage devices, typically spinning disk drives, that are located inside a server, also known as the *filer*.
- ✓ A user's computer attached to the storage via the network that sees files and folders just like it does on a local hard drive except that this folder may have other users' files in it as well.

One of the nice things about NAS is that depending on the platform, tens of thousands of clients can be attached to the shared storage areas, all happily reading and writing the files they store there.



The ability to share data easily is fundamental to collaboration and efficiency in a workforce.

What made NAS grow so quickly and easily is that the basic protocol, or language, that it uses to communicate isn't proprietary, meaning that you don't have to have a particular operating system, server, or storage vendor to make it work. The NAS protocol comes in two types, but both provide the same feature set to allow shared access to files across a network from different client computers simultaneously:

- ✓ **Network File System (NFS):** This protocol was originated and adopted as a standard on Unix-based operating systems.
- ✓ **Server Message Block (SMB):** This protocol came about later. When Microsoft came out with the Windows operating system, the company decided to adopt its own network file protocol. SMB is widely referred to as CIFS (Common Internet File System) in the NAS industry, and for this reason, I use SMB/CIFS in this book to refer to this protocol.

Both of these protocols are widely supported and formally standardized across most operating systems and NAS vendors. A standardized protocol for NAS enables users to

choose and share information without worrying about compatibility issues with Linux, Windows, or Mac-based client operating systems. Users can select the best hardware and software for their business.

Handling the Incredible Growth and Management of Data

Data comes in two basic types: structured and unstructured. *Structured data* has a definite format, such as what you would find in a database. Each record, field, and data type expressed within the file follows a predefined layout. What's nice about structured data is that you can describe every bit of the data and explain what each byte of the file is. You can jump to any spot inside a structured data file and by following the predefined layout as if it were a map, you would know what the bytes you're looking at actually represent — for example, a database that contains customer records with names, addresses, account numbers, and most recent purchase dates. You can, if you want, delete only the records of customers who haven't purchased anything for more than a year, which would shrink the size of the database and save you space. The structured layout makes it easy to find the records you want to eliminate.

Unstructured data has no practical internal definition. Adding or subtracting data from a file that is unstructured is much more difficult than doing so in a structured file. You *can* do it, but it's not as easy as just running a database query to add or delete certain records, as you can do with structured data. Unstructured data usually takes the form of such things as documents, spreadsheets, and *data streams*, such as audio, photos, and video files. The application that created the file knows how to deal with it, but in terms of the bulk management of unstructured data, you are at the mercy of the sheer number of possible applications that deal with these files correctly.

Over the last few years, the type of data created and the size and amount of data that companies have kept on hand has continued to skyrocket, and unstructured data is the biggest offender in this growth (see Figure 1-1).

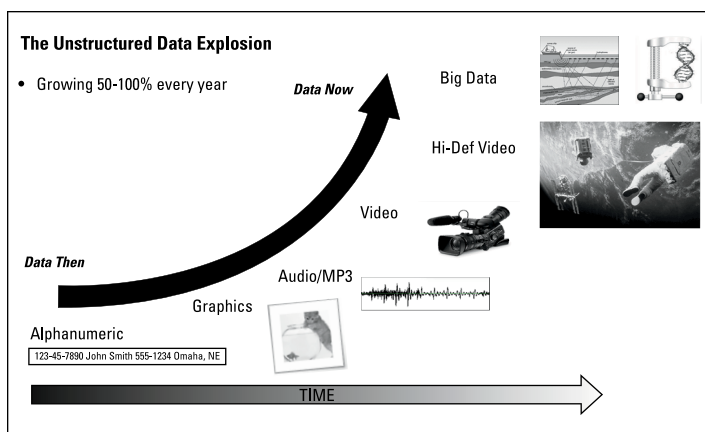


Figure 1-1: Unstructured data is growing more plentiful and bulky to deal with.

You may have witnessed this yourself by building more and more NAS systems to hold all this data. All these “NAS-islands” need to keep up with not only the physical storage of all this stuff but also with the demand for client access. Apart from getting a call from someone complaining about a full NAS share or slow performance, how do you gauge your storage efficiency and limitations? I talk about that in the following section.

Finding an Efficient Solution to Overloaded NAS Systems

When you have all that data to manage, you have to manage it wisely. Data that was important yesterday may not be important today — or it might be even more important now than it was then. You never know how business needs will change the value of your data, so dynamic data management is crucial to adapting to changes. NAS storage has the great advantage when compared to traditional block-based storage of being more mobile and, therefore, can provide intelligent data placement. The network that NAS storage is attached to gives you that flexibility to move your files around to wherever suits them best.

Having a layer of technology above all your NAS solutions that recognizes access patterns to files and can shuffle and redistribute files according to changing business needs would be an incredibly convenient tool to have. I talk about what that solution is later in this book.

Chapter 2

Understanding Cloud

In This Chapter

- ▶ Discussing the cloud's pluses
- ▶ Understanding that the cloud isn't a perfect world
- ▶ Looking at the differences between legacy computing and cloud computing
- ▶ Talking about the cloud vocabulary
- ▶ Explaining what makes up the cloud

The term “cloud” came about from IT folks doing diagrams of their servers and networks to show how everything is connected; when they didn't know exactly what to draw for a certain section, they would just draw a cloud-shaped symbol. When the Internet became part of the connectivity, IT people didn't know what “the Internet” equipment was either, so they would draw another squiggly bubble, resembling a cloud, to represent “we're connected to something inside this mysterious cloud.” In other words, the cloud is really just a metaphor for the Internet itself. You don't know what you're really talking to, but you know it's there.

This chapter covers what cloud computing and cloud storage look like and discusses the similarities and differences between the cloud and the traditional server, storage, and network infrastructures you may be accustomed to. It also goes into the reduced cost of using cloud services and how you can use the cloud to augment your current computing and storage platforms securely and easily to save money, space, and headaches.

Appreciating the Cloud's Advantages

A significant reason that major companies have adopted cloud is that the gear to run the cloud and store the data doesn't reside at a company's physical site, which means that the company isn't building computer rooms, buying gear to fill them, and maintaining that gear over its normal lifecycle. Think about it like a rental car agency. Rental car agencies buy the cars, house the cars, and maintain the cars. You just pay them to use one whenever you need to.

Cloud works on a very similar idea. Cloud sites build the car, and you rent the time to use it. When you're done and you return the car, you pay for the time (and gas) and that's all. With the cloud, you can rent time on compute servers or rent storage space so quickly, and on systems that are so fast, that it would be impossible for your own staff to design, purchase, and install your own storage and computing infrastructure in the same amount of time. Cloud gives you that quick and cost-effective turnaround in setup, use, and eventual return of computing resources.



Cloud services have another huge benefit. Because of the scaled size of the resources that provide the computing and cloud storage, the price you pay is incredibly cheap compared to what you would pay to build it, power it, cool it, and maintain it yourself. And because money is usually the driving force in most IT decisions, and storing data inside the cloud is very cost effective, this is actually the biggest cloud benefit.

Ah, Yes — the Cloud's Disadvantages

Every coin has two sides. As discussed in the preceding section, the cloud has huge advantages. Unfortunately, though, the cloud also has things that make people feel skittish. The cloud is outside your normal data center walls, which means that the gear that's running it isn't under your control. Also, the immediate physical security and long-term data storage security is now someone else's domain.

When someone else controls your storage and computing environment, you may experience a little anxiety. However, cloud service providers know that they have to deal with that problem. Many, if not all, cloud providers protect their environments by having no single point of failure, physically and geographically. In other words, your applications living in the cloud are actually capable of running *anywhere* and, most likely, *everywhere* at some point. In cloud services, multiple physical data centers that back up each other and protect the physical equipment are the norm.

Protecting the data is a concern as well. Your organization is transmitting data out of your data center and into the cloud providers' network and servers. Sure, security is in place to make sure that your data is protected, but some people are a little queasy about not having complete control over their own environments. You *can* secure your data within the cloud.

Learning the Lingo: Cloud Terminology

Cloud computing resources are, in essence, the same as those of your typical data center:

- ✓ Servers that your applications run on
- ✓ Storage devices where your data is written to and read from
- ✓ Networking resources that connect the servers, storage, and users who exist outside the cloud to your applications

Pretty simple, right? The cloud isn't that much of a mystery. Once you get the lingo down, you'll be far more comfortable with using the services that the cloud provides. Cloud servers and networking are relatively the same with just a few nuances thrown in, which I discuss later on. The major difference with the cloud is how you store data, which is a good thing because it makes things cheaper and much more flexible.



Storing data in a cloud, specifically with a public cloud solution such as Amazon S3 (or Simple Storage Service), is just one way of using the vast resources of the Internet to store and retrieve objects. The reason I say *objects* instead of *files* or *data* is because the concept of an object is the atomic structure that has to be fully understood before any other details of the cloud can be discussed. But, before I go into what objects are, I need to define two other terms: namespace and buckets.

Namespace

A *namespace* is a way to organizationally manage many objects within a big container. Namespaces can span physical locations and systems. A namespace contains uniquely named items, such as buckets, which I talk about next. Namespaces help to fence off a set of unique objects from other uniquely known objects, which exist in other namespaces.



Think of a namespace as a universe. Each universe knows about everything on its inside, but it doesn't know or care about anything outside its realm.

Buckets

Buckets are located within a namespace. A *bucket*, which I think is a really great name, is a portion of a namespace that is totally unique in name within a namespace.



Buckets contain objects, which are the equivalent of a file. To refer to an object, you need to know the namespace, bucket, and object name, all of which are functionally equivalent to a directory-path/filename combination.

Objects

Objects are the actual data that you store and are all unique items. Objects are addressed by something called a *key-value pair*, which has two halves, the first half of which is essentially a filename, called the *key-name*, that exists only once within a globally defined namespace, inside a bucket. The *value* part of the *key-value pair* is the actual data within the object. For example, here is how an object is referenced:

`http://s3.amazonaws.com/bucket/key-name`

The link includes

- ✓ The *http://s3.amazonaws.com* portion, which is the web address of the Amazon S3 namespace.
- ✓ The *bucket*, which is an unimaginative unique name for a bucket.
- ✓ The *key-name*, which is the object's name and represents what would, in a standard file system, be a filename.

In a more detailed example, say that you have an Excel spreadsheet that is your company's ("XYZ Corp") quarterly earnings report that you want to save to the cloud. Here's an example URL to reference that file, or object, in the cloud:

```
http://s3.amazonaws.com/XYZ_Corp_Data/  
2014_Q2_Earnings_Report.xls
```

That's pretty straightforward, isn't it? XYZ_Corp_Data is your bucket, and everything after the bucket is the object ("2014_Q2_Earnings_Report.xls")

What's in a Name?

Here's the catch. In standard file systems, you have directories and subdirectories (or folders) to help organize data in some hierarchical fashion. Inside the cloud, you really don't have an exact concept of a folder or directory. Everything is addressed with Bucket/Object names only, not Bucket/Folder/Sub-Folder/Sub-Sub-Folder/Object. Amazon S3 allows you to cheat this lack of concept by using "/" in the key-name of the object. So, for the Excel spreadsheet example provided earlier, you can do this:

```
http://s3.amazonaws.com/XYZ_Corp_Data/  
Reports/Public/2014/Q2/Earnings/Report.xls
```

In this URL, which is significantly more complicated, the bucket is still XYZ_Corp_Data since that is always the first thing after the namespace (*s3.amazonaws.com*), but the remainder is still the key to the object, which in this case is the Excel spreadsheet for your 2014 Q2 Earnings Report. If you go to Amazon S3 using their web-browser-based administration interface and look inside the bucket, you'll see folders under the XYZ_Corp_Data bucket that you can click on to drill down inside them like in a normal directory or folder, until you get to the actual

“Report.xls” object. However, the key-pair for that file is, and always will be, “Reports/Public/2014/Q2/Earnings/Report.xls.” You can’t refer to it only as “Report.xls” in any sense because “http://s3.amazonaws.com/XYZ_Corp_Data/Report.xls” and the original URL refer to two different bucket/key-pair names entirely.



Be careful how you think of traditional filenames and object names within the cloud. No more is there a referential view of being inside a folder and then pointing to the files inside that folder and only that folder. It’s also important to remember that there is no concept of a directory or folder, as in a standard file system, and an object doesn’t actually exist inside or under a folder. Amazon S3 allows you to pretend that a folder structure exists, but there really isn’t any such concept. The object’s key-name is a fully qualified name from the bucket name down, and you can’t rename these folder names within the key-name and expect any other object that shares that same folder name to also magically move to the new folder — because there isn’t a folder; what looks to be a folder name is actually part of the object’s name. You would have to rename every object’s key-name one by one in order to change the pseudo directory/folder representation.

You must be specific every time you describe an object, from the bucket all the way down to the object key-pair names. You can translate your current methods of storing and retrieving data in the manner you’re accustomed to into the cloud’s object-based storage scheme. Avere Cloud NAS does this transparently for you as part of their data-tiering capabilities when it stores data in the cloud.

Parts of the Cloud

A misconception exists that the type of equipment that makes up the cloud is different than standard computer gear that you would use in your own company’s data center. This isn’t the case. The cloud uses the same server, storage, and networking gear that any regular IT shop uses — but it’s the way it uses them, and what they do, that makes the difference.

The basic cloud ingredients are

- ✓ Compute servers, which run an OS such as Windows, Linux, and so on
- ✓ Storage servers, which hold your files
- ✓ A network, for communication

Compute servers

Compute servers are where the application magic happens. These servers read the data, crunch the data, and then spit out information. Servers in the cloud tend to be grouped into a *compute farm* or a multiserver pool of resources consisting of CPU, memory, and sometimes limited local disk storage. This pool is “virtualized” into one super-server, and a virtual server is created by slicing off X number of CPUs, Y number of memory, and Z number of virtual disks.

Many of these virtual farms are running VMware or Xen virtualizing software to do all the slicing and dicing of resources. Think of a big loaf of bread being the pool of all the servers’ CPUs, memory, and disks, with a slice of bread from the loaf as your new virtual server that behaves like a single, physical server. Making all the resources of the server completely virtual allows a system administrator to move the operating system and applications from one physical server to another on the fly, without interruption.

When you request a server to be created in the cloud, you can describe what operating system you want it to run, how much memory you need, and the CPU count. You also can choose from varying types of standard setups, depending on what kind of applications you’re planning to run.

Storage

Although it can vary, *cloud storage* is always a type of shared storage that can be connected to multiple servers simultaneously. This ability to share makes it easy to slice off what you need and give it to whichever server needs it. When you’re done, you can even disconnect it and give it back to the pool for other servers to use again.

Storage in the cloud can have different characteristics, such as redundancy and a guaranteed I/O rate. You'll pay more per gigabyte for highly available and/or faster storage, which is a choice you make depending on what you need when you request the storage resources through the cloud administration tools.

Network

The glue that makes everything work is the network. Cloud networks are *virtual*, and they're great because you don't have to run around plugging and unplugging wires to get servers and storage and clients to talk to one other. The servers are connected to the network when they're built, and everything else, such as IP addresses, routers, DNS entries, and firewall rules are all handled either automatically or customized with management software by the administrator.

Looking at Cloud Types

Clouds come in two basic "flavors." You can build your own internally functioning cloud, which is called a *private cloud*, or you can use a publicly accessible cloud solution called a *public cloud*. The cloud also has a third type, but instead of a flavor, it's more of a mix known as the *hybrid cloud*, which is just a private cloud and a public cloud connected together that pass data back and forth as needed.

Private clouds

Private clouds are storage and computing environments that you would build yourself, except that all the interaction between them and your applications would happen in *cloud-ese*, or the protocols that cloud computing is accustomed to speaking. For example, you can use TCP/IP addresses exclusively to communicate with applications, use object-based references to data, as discussed previously, and have web-browser-friendly, client-side applications.

Private clouds function in the same way as public clouds except that all the hardware is inside your own data center and not out on the Internet being shared with everyone else. A private cloud places everything under your control, but the downside is that you have to build it and maintain it.

Public clouds

Public clouds, which are the more prominent cloud services solution, exist in the Internet's shared space. When so many different companies share the resources, the price of compute and storage on the public cloud goes way down. You pay only for what you use, which is perfect for companies that have sudden demands for computing resources such as a product launch, a blockbuster movie to complete, or a big genomic study. When the demand for resources goes down, or is no longer necessary, the resources are returned to the public cloud for others to utilize, and you no longer pay for them.

Hybrid clouds

You don't have to pick one type of cloud over the other though because the *hybrid cloud* is a mix of using together both private and public cloud services, which gives you the control of in-house private cloud resources plus the cost savings and flexibility of public cloud. Most use cases involve using private cloud to do day-to-day data processing and then storing that data to the public cloud once it becomes less useful but is still important enough to keep. Because the private cloud's network links are usually higher speeds and don't require any transfer costs, storing data in the private cloud is better for data that you access more frequently. If you use the data less frequently, then sending it out to the public cloud is usually a more cost-effective way of storing it for the long term.

Note: Moving data from within the company's private cloud to the public cloud isn't a trivial operation, but using Avere Cloud NAS to bridge the gap between them is an efficient way to do so.

Accessing and Securing the Cloud

How you communicate with cloud resources is simple. If you're using a private cloud, which is inside your own corporate network, the cloud would appear in the same way that any other application server does. A public cloud solution isn't that much trickier.

Once you have TCP/IP connectivity, the cloud needs to confirm your identity before allowing you to access your servers, namespaces, buckets, and objects. Security is very tightly controlled in every public cloud solution.

Speaking generally about most cloud security, you'll have specific operations that can be performed on certain components within the cloud. These abilities or *roles* are usually assigned or revoked by user or group of users. Each user has a specific set of authentication/access keys that have to be verified every time they communicate with the cloud provider. These keys ensure that the application (via a user interaction) is a legitimate cloud services user and can perform the tasks requested by the application. Basically, the security of cloud resources is as good as, if not better than, your current internal IT infrastructure. Security is everywhere in cloud, from the access to the servers to the communications between them and all the way down to the objects placed in the buckets. Everything can be controlled and monitored.

By using a cloud solution such as Avere Cloud NAS, in addition to normal user- or group-based access, you also have the ability to encrypt the data that is stored on the public or private cloud. Doing this ensures that nobody can access the objects (files) you've stored there, so they're safe from unauthorized eyes. Only Avere filers can decrypt and access the data held within, which helps to ease the worry of storing important proprietary customer data on the Internet.

Transitioning into the Cloud

Now, you're probably thinking, "Okay, so how do I make this happen without throwing out my existing gear?" Don't worry. The Avere Cloud NAS solution will help you optimize the investment in your current NAS environment while gaining all the benefits of cloud discussed here. Chapter 3 explains exactly how to bridge the gap between the cloud and NAS to create Cloud NAS.

Chapter 3

Avere Cloud NAS

In This Chapter

- ▶ Combining your NAS environments with cloud storage
- ▶ Moving data the easy way
- ▶ Taking good care of your data
- ▶ Utilizing the strengths of both NAS and cloud storage

Chapters 1 and 2 talk about NAS storage and cloud storage, respectively. This chapter discusses how to take the pros and cons of using traditional NAS and combine them with the pros and cons of moving to cloud storage to make a best-of-both-worlds solution called Avere Cloud NAS. Combining both ways of storing your data takes advantage of all the positive benefits of NAS and the cloud and mitigates the risks of all the negatives. The whole really is greater than the sum of its parts!

Bringing NAS and the Cloud Together As One

NAS has been around a very long time and has the advantages of easy connectivity and management. The downside is that in today's world of exponential day-to-day data growth, NAS just isn't able to keep up with increasing business requirements. That's not the fault of NAS itself; it's the scale of the problem and the dynamics of change that are really the big issue. Because data and processing power increases, changes scope, and evolves over time, foreseeing — let alone building — the perfect NAS environment for tomorrow, next month, or next year is simply impossible. Every NAS solution that's been built

to meet the demand of a Day One rollout became obsolete as soon as it was turned on. Why? Because things change. In no time at all, that shiny new NAS solution you installed is just another box that will soon overflow with bits and bytes of who knows what data. You'll start planning the next NAS solution as soon as the relevant business data your users are storing starts to age, becomes "cold," and rarely gets referenced. New files with relevant data are piling up fast!

Cloud storage, on the other hand, is designed to deal with changing requirements. Cloud storage expands to fill the need of the files you store there and also makes for a cost-effective place to store data that has become obsolete but can't be completely trashed (check out Chapter 5 for more about archiving to the cloud).

By taking advantage of both types of environments simultaneously, you would have a win-win situation for your data storage needs. This is where convergence of the two solutions comes into play and makes a lot of sense. Cloud NAS is where bloated legacy NAS solutions and efficient, economically scalable solutions of the cloud complement each other perfectly.

Avere Cloud NAS isn't a replacement of your existing NAS infrastructure. Cloud NAS is the resulting combination of Avere FXT intelligent NAS filers together with your already-in-place NAS platforms.

Take a look at Figure 3-1. The left panel shows a traditional NAS setup before Avere FXT filers are added to the mix. Traditional NAS environments face numerous challenges in today's world, primarily related to scaling performance, keeping cost under control, managing data spread across many silos of storage, and providing global access to shared data.

In the middle panel, Edge filers from Avere provide NAS Optimization and transform the monolithic NAS pools into an "edge" and "core" design. This design tackles the problems of trying to scale-out your NAS environment for increased client throughput while also enabling intelligent and efficient data placement to reduce the costs to store it all.

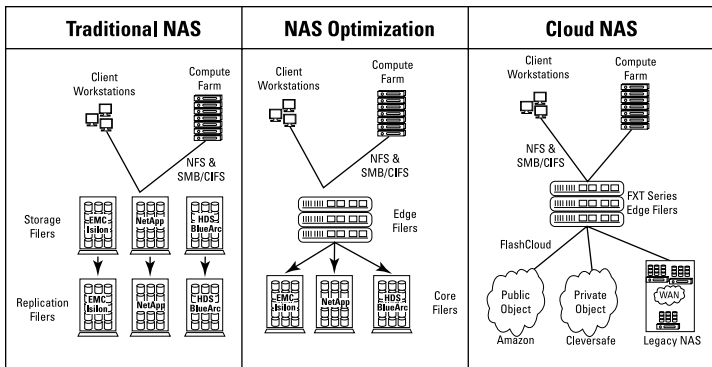


Figure 3-1: Avere Edge filers transform traditional NAS into Cloud NAS by optimizing the Edge and Core.

As the middle panel of Figure 3-1 shows, the Edge filers from Avere are now the component closest to the client's and compute farm servers, compared to the traditional NAS. In Avere Cloud NAS, groups of Avere Edge filers take over this file-serving role from existing NAS filers. Avere Edge filers are now responsible for servicing all your client and compute farm I/O requests directly. You still store existing files on their native Core filers, so to take advantage of Cloud NAS, you don't need to migrate data. Because the Avere Edge filers contain internal fast drive technology to handle file read, write, and metadata operations locally, the high-speed internal cache vastly speeds up I/O for clients. As files are written into the Edge filer's local cache, the least recently used (LRU) blocks are transferred in the background to the cheaper, slower, long-term storage of your Core filers.

The Core filers contain the longer-term storage for files to live in. Existing data in your NAS environments starts at the core when you overlay it with Avere Edge filers. The Avere Edge filers are now performing the file-sharing services, presenting your clients with the NAS shares they're accustomed to. As portions of frequently used files are accessed from Core filers, those portions of the file are copied into the Edge filer's cache for quicker access times. Conversely, when data is less frequently used and becomes cold, the data gets demoted and moved away from the Edge filers onto the Core filers for more feasible lower-cost-per-MB storage. Moving data around based on usage and cost is known as *data tiering*.

In the third panel in Figure 3-1, you see that the Avere Cloud NAS adds both public and private clouds as available locations to store data apart from legacy NAS. Avere Cloud NAS extends the performance, cost savings, simplified management, and global access benefits of NAS Optimization into the cloud. Performance can be scaled to millions of IOPS and hundreds of gigabytes per second (GBps) for data stored in the cloud. Public and private clouds provide the most cost-effective storage available, and Avere Cloud NAS integrates these into a global namespace together with your existing NAS. Avere Cloud NAS also provides global access to data with the cloud being the perfect centralized data repository.

Upon adding Avere filers to your environment, your legacy NAS environments are not altered and your current data continues to reside right where it always has been. What Avere Edge filers provide is the ability to enable data tiering and seamless cloud storage accessibility to your environment with a simple mechanism for migrating data between platforms. Avere's internal algorithms detect file usage activity and react to move data to where it will be most efficient — to the Edge or in the Core.

Using the Best of Both NAS and the Cloud

The old saying that “two heads are better than one” really comes into play with Cloud NAS. With the Avere Cloud NAS Edge-Core architecture that leverages NAS and cloud storage, you can solve the storage problems that traditional NAS and general cloud storage can't fix independently. Speed, cost, and complexity are the main issues with managing storage. By applying the features that make NAS or cloud a good solution individually, you can use these features to solve each issue in turn.

Making things faster

The Avere Edge-Core architecture separates performance from capacity and delivers both more efficiently. This architecture takes a typical NAS solution and turns it into two parts: the

Edge filer and the Core filer (see Figure 3-2). The Edge filer, optimized for performance, is what makes things faster.

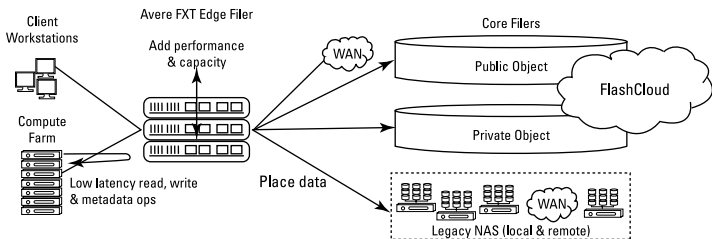


Figure 3-2: Avere Edge-Core architecture.

The Edge filer provides high-performance and low-latency data access to the users. This is accomplished by three distinct features of the Edge filer:

- ✓ Dynamic tiering algorithms that run inside the Edge filer automatically move the most active data blocks from the slower disks of the Core filer(s) to the Edge filer's faster RAM, SSD, or SAS drives.
- ✓ The Edge filer, a full-fledged file server, locally terminates all read, write, and metadata operations. I/O is cached back to the Core NAS silo to reduce latency.
- ✓ Multiple Edge filers, starting at three and going up to fifty, are clustered together and work as a group, distributing file requests from clients and dynamically scaling the I/O workload across themselves.

With an Edge filer on the front lines handling all the performance of “hot” data, the demands on the Core filer(s) are reduced and can be optimized better for capacity and cost savings. The next section takes a closer look at where the cost savings enter in.

Making things cheaper

Data has value, and storing data has a cost. Data also has a lifecycle, or in other words, the degree of usefulness that data has changes over time. Every line of business and every

type of data in that business has a varying lifecycle curve. To be 100 percent efficient with your data storage, you would need to track the usefulness of every piece of data throughout its lifetime. Because doing that isn't practical, the better approach is to let the activity of the files dictate how important they are. The more active files have frequent access while files that sit idle for long periods of time take up space that may be needed for files with more important or more valuable data. By matching value of data with cost of storage, you save money by offloading irrelevant, less important data to slower, cheaper-to-use storage, and you make room for higher-value data on high-speed storage.

With the Avere Edge-Core architecture, the FXT Edge filer cluster provides a buffer in front of the Core filers. This buffer absorbs the performance workload, insulating the Core filers for the I/O and bandwidth requests from the client computers and application servers accessing the storage. By offloading the performance demand, the cost of the Core filers can be reduced. You can accomplish the cost reduction by using less-expensive storage controllers and higher-density and low-cost drives.

Avere solutions typically require less total equipment than traditional NAS deployments, resulting in dramatic savings in storage capital expenses and ongoing operational expenses for power, cooling, and rack space. With Avere in place, the total cost of ownership (TCO) of a NAS environment is reduced. (See Chapter 5 for more about reduced costs.)

Making things simpler

Avere offers global namespace (GNS) functionality that simplifies data management and access in storage environments that have multiple storage systems. Avere's GNS spans public object, private object, and legacy NAS, all from heterogeneous vendors (see Figure 3-3).

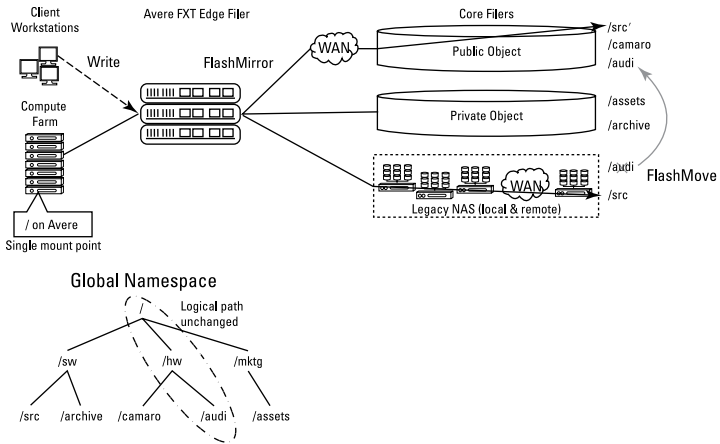


Figure 3-3: Public object, private object, and legacy NAS in a global namespace.

With the Avere GNS, clients have a single mount point on the FXT cluster and can access all the data across all the systems.

Note: GNS solutions have gotten a bad rap from older generation solutions that had a negative impact on performance or intrusively remapped the structure of your file systems to the point that they caused confusion and became impossible to remove. Avere provides GNS and performance acceleration in the same product, and the Avere FXT cluster is simple to add to and remove from existing environments. The clustered FXT filers simply share the mapping of the GNS structure you create, and that virtual map points to your existing Core filer systems, including public and private cloud connectivity. The cluster of filers access the Core and cloud data silos no differently than you would natively — so you're not trapped into using GNS forever by any vendor lock-in.

With the GNS in place, you have a single map to all your data repositories and can now provide transparent data migration using another built-in Avere FXT filer feature, FlashMove software.

Easy Data Migration

Avere FlashMove software running on an Avere FXT cluster takes the pain out of NAS data migrations. Behind the FXT performance tier, FlashMove nondisruptively moves live data between Core filers, public object, or private object stores. This enables easy-to-implement, nondisruptive load-balancing and archiving capabilities and simplifies the laborious task of retiring older storage silos when transferring existing data to newer, more supportable systems.



As time passes and your business grows, data migration becomes a necessary evil. More users, new applications, and faster server processors accessing your data push existing storage systems to their performance and capacity limits. Data migration is necessary to rebalance the storage, but traditional methods lead to user disruption, application downtime, and reduced productivity. Some specialty data migration solutions exist, but these are costly, require complex planning and risk, and can have a negative impact on performance during the transfer from one system to the other.

With FlashMove, you don't need to halt applications or suspend access to data during migrations. FXT clusters serve active data to application servers and users while FlashMove software transparently moves data between Core NAS filers. Managing your NAS environments is simplified. You can load-balance live data across existing systems, transparently archive to secondary storage, add new storage and new vendors to the NAS environment, and decommission old storage that is no longer useful.

Protecting Your Data

Migrating your data from one location to another is accomplished via the FlashMove software feature, discussed earlier. But what if you need to make a mirror image copy of your data? There's a way to do that too. Avere FlashMirror software running on an Avere FXT cluster provides this. FlashMirror is a flexible and high-performance solution for implementing comprehensive data protection across a NAS infrastructure. Integrated with the native tiering on the FXT cluster, FlashMirror quickly and efficiently replicates data to multiple NAS filers, even heterogeneous ones made up of multiple vendors. FlashMirror is easy to deploy and manage, offloads

replication processing from the Core NAS filers, and scales replication performance to any level required.

Enterprises require continuous access to critical data to keep the business running. Failures, such as an equipment failure, a power failure, or a severed communication link, are inevitable and must be guarded against by the enterprise's Disaster Recovery (DR) process. Typical DR processes include replicating or mirroring critical data to a secondary site. Implementing such a process is expensive, difficult to manage, burdens storage performance, and does not work well between different vendors' equipment.

Avere FlashMirror simplifies the implementation of a DR practice on a NAS infrastructure and replicates data on primary and secondary NAS filers. It also keeps them in sync by sending updates directly and in parallel to both filers. FlashMirror offloads replication processing from the storage and supports clustering to scale replication performance to any level required.

Connecting to the Cloud

Enterprises understand the improved economics and technical advantages of object storage for storing large quantities of file-based, unstructured data. When compared to traditional storage approaches, object storage offers lower equipment costs, lower operating expenses, simplified management, and reduced facility footprint. And, with massive scalability, simplified management, and built-in redundancy, object storage provides a great technical solution for large-scale file stores. With limitations such as high latency and an unfamiliar object-based access protocol, however, enterprises face challenges to widely deploy object storage.

Avere FlashCloud on FXT Series Edge filers integrates public and private object storage with legacy NAS into a global namespace and provides scalable performance for users everywhere via familiar NAS protocols. GNS gives enterprises the flexibility to store their data wherever it makes most sense and adopt object storage at a comfortable pace. FXT Edge filers provide enterprise-class NAS functionality protocols, including NFS (Network File System) and SMB (Server Message Block), which are widely referred to as CIFS, the Common Internet File System (refer to Chapter 1), scalable

performance, and high availability to support any applications running on NAS. Together, FXT Edge filers and object storage provide optimized performance, capacity scaling, and increased TCO savings.

Another benefit of Avere Cloud NAS is the ability to integrate directly with cloud storage without all the complexity around dealing with the cloud's naming conventions, security, and access methods. Because you interact with cloud storage completely differently than with traditional NAS filers, to utilize cloud storage directly would require new application interfaces and user training on how to access those files. With FlashCloud software, the complicated cloud storage interfaces are hidden, and your files look just like they normally do within your NAS environments.

Making Things Better

What does better mean? Better means all the virtues described throughout this chapter: faster, cheaper, and simpler. Better also covers operational resiliencies such as failures in componentry, network links, power, and the cooling systems within your data centers and across the globe. The Avere Cloud NAS filers provide a feature called high availability (HA), which means that they work together as a cluster and are designed to cover for each other in case a filer or network link fails. Having the peace of mind that your NAS infrastructure has built-in resiliency in addition to scalability definitely makes things better.



Cloud NAS merges the long-standing connectivity and easy use of NAS with the future direction of data storage. Along with that, scalability, investment protection, and consolidated management make the Avere Cloud NAS solution a better place for your enterprise.

Chapter 4

Cloud NAS for Big Data

In This Chapter

- ▶ Addressing the challenges of Big Data in various industries
- ▶ Retrieving Big Data
- ▶ Seeing how Cloud NAS technology can solve Big Data issues

Big Data sounds like something that only Texas has, but Big Data is actually very common in industries that have applications with very heavy compute requirements for huge amounts of data. Because of the sheer amount and complexity of Big Data, the ability to serve it up, manipulate it effectively, and extract critical business-decision-making information from it is a make-or-break scenario for businesses that rely on manipulating humongous data sets.

In this chapter, I discuss which industries have Big Data compute and storage needs and why those needs are tough to deal with. I also show you how Avere Cloud NAS makes using Big Data much easier for your enterprise. By seeing some real-world examples of how Big Data is used, you'll come to understand that Big Data isn't something to fear.

Note: Similar to the term “cloud,” many complex and varied definitions exist for Big Data. For the purpose of this chapter, Big Data is defined as a huge amount of data that needs to be analyzed to extract valuable information. That shouldn't be too difficult, right?

Crunching Big Data in Media and Entertainment

Those exciting, edge-of-your-seat blockbuster movies that you see are a result of Big Data. Rendering can make the difference between a box-office miss and the blockbuster movie that packs the house with its spectacular scenes and heart-stopping sequences. Every filmmaker strives for such a compelling impact: creating a space shot so realistic that when the astronaut's tether breaks from the space station, the audience gasps with him, or constructing a screen-flooding wall of water so real that moviegoers catch their collective breath when it recedes. But producing such gripping scenes requires massive high-speed storage resources to manage the huge amounts of data — and studios have to balance visual impact with the earthly realities of tight budgets and production schedules.

Avere Systems storage solutions make it possible for visual effects production (VFX) teams to deliver more visual impact in less time and at a lower cost than ever before. Avere FXT Edge filers offer linear performance scaling to millions of operations per second and more than 100GBps (gigabytes per second) throughput at a capital equipment cost that is two-thirds less than traditional NAS solutions. The Edge filer automatically places active data — such as texture files — on solid-state media to meet the demands of complex rendering workloads. Avere solutions deliver performance for hundreds to thousands of render nodes while ensuring maximum responsiveness to artist workstations.

So, the next time you watch a movie, check out the texture of that wet monster's fur and where each water droplet goes when he shakes his body and think to yourself, "Yep, that's Big Data."

Crunching Big Data in Oil and Gas

Of all the applications in the oil and gas industry, seismic processing places the greatest demands on computing and storage. Geoscientists conduct huge surveys of the earth's

subsurface collecting sound wave data from areas that might have oil and gas reservoirs. Taking these seismic snapshots of the ground can cost hundreds of thousands of dollars per square mile. These data files and the process used to develop these pictures into usable exploration maps is the role of Big Data.

That Big Data is processed using many “what if” scenarios to decide if and where to drill. Researchers are stuck in a tricky place. They must provide fast throughput of petabytes of Big Data while managing compute and storage costs. Avere’s Edge-Core architecture separates performance scaling from capacity scaling and more efficiently delivers both. Avere Edge filers are optimized for performance and scale throughput for seismic processing to more than 100GBps. By streamlining the end-to-end upstream (seismic processing, seismic interpretation, and reservoir engineering), workflow geoscientists can achieve faster time to final results and get that gasoline to the pumps quicker. So, the next time you’re pumping gas and want to think about something other than a lighter wallet, think about Big Data.

Crunching Big Data in Life Sciences

Science is about big ideas that often depend on Big Data, which in turn requires big storage infrastructures. Getting the most from that data now requires enterprise architecture that can come at a big price. Avere speeds your time to results, helping to meet demanding publication and evaluation deadlines.

The human body is an amazing structure of complex systems. Each system is encoded with data that governs how it operates, and that data can be captured through genetic-sequencing processes. Hiding somewhere in the billions of nucleotides sequenced at labs and hospitals around the world is a wealth of genetic data that can improve the outcome of medical treatments, help cure the most intractable of diseases, and reduce the costs of healthcare. That’s incredibly valuable Big Data, but to fathom the complex mysteries of DNA, genomic

researchers need the processing power of large compute farms and Big Data storage that can handle the heavy read/write workloads of sequencing — without adding undue IT cost and complexity.

Avere storage solutions help researchers produce and analyze more data faster and at a lower cost than traditional NAS systems. The Avere FXT Edge filer automatically places active data on fast RAM, Flash/SSD, and SAS media to optimize performance and ensure maximum responsiveness to genomics and other life-sciences applications. The clustering enables linear performance scaling to millions of operations per second and more than 100GBps throughput.

Crunching Big Data in Software Development

Fast-paced software development in high-growth businesses leaves many successful software development companies struggling with software tools repository distribution and performance. As development grows, it can get out of control. Development teams from around the globe collaborate 24/7, creating Big Data sets to manage. Avere Edge filers speed access to the latest testing and development tools and keep the development process moving without the high cost of traditional NAS hardware and replication software.

The fast-paced nature of the development industry leaves high-growth companies with heterogeneous storage environments that are difficult to manage. New projects can spring up out of nowhere, and several branches of the same development cycle can occur in parallel, multiplying access requirements. Administration of multiple file stores across various storage environments for these newly requested data areas can slow the development process.

Global namespace gives you a single view of files stored on both legacy NAS and cloud-based systems, simplifying the administration and accessibility to your developer community. Unlike other global namespace offerings, Avere lets you keep your storage options open by supporting all the common traditional NAS suppliers *and* object storage as well.

Crunching Big Data in Electronic Design Automation

Everyone loves gadgets — from smartphones and tablets to home automation and automotive media-control systems. To create these gadgets, electronic design automation (EDA) tools leverage storage, making performance critical. Engineering teams from around the globe work together to analyze and perfect the latest gadget (so that you'll just have to have it). Such a high-performance design environment can easily be misconfigured, resulting in instability and bottlenecks. The mass production of electronics that spread across the globe in the blink of an eye has to be perfect the first time. A mistake in design that produces glitches, bugs, and failures can ruin a company's reputation. Many of the world's best EDA companies have secured performance in their NAS infrastructures by adding Avere FXT Edge filers in front of their core storage hardware. Edge filers help eliminate latency and optimize performance to support the design environment.



As development grows, storage can get out of control. Scaling storage performance and capacity cost effectively is difficult. With Avere FXT Edge filers, developers can get both. Scale-out clustering scales performance with up to 50 FXT nodes in a single cluster, ensuring high availability of data even in the event of failures. Capacity of the FXT local fast cache scales up to 450TB (terabytes) — which is the largest local cache capacity.

The fast-paced nature of the EDA industry leaves high-growth companies with heterogeneous storage environments that are difficult to manage. Avere Global Namespace gives you a single view of your existing file stores on both legacy NAS and cloud-based systems. Because the EDA process never really stops, to install faster, more-scalable storage solutions, you need to do so without disruption. Integration of Avere filers into your environment is easy to accomplish without disrupting ongoing processes in your current environment.

Crunching Big Data in Web Hosting

Hosting 500,000+ websites or millions of digitized archives is again Big Data. Users have come to expect instant access to these huge repositories of information with just a few clicks. Staying ahead of consumer demand and expectations isn't easy. Managing the hosting websites, media, or on-demand content requires a flexible storage infrastructure and a little bit of ESP. Whether you're serving websites or the latest blockbuster movie, storage performance is a key to consistent on-demand access that creates an experience that users will want to keep coming back to.

Avere FXT Edge filers sit in front of your Core storage and automatically identify what data is hot and what's not, pushing the most-used data to the highest-performing storage media in the Edge filer cluster as close to your customer's browser clicks as you can get.

Accessing Big Data

As you've no doubt gathered, Big Data requires some hefty resources in storage and computing to make it useful. Today's general file-sharing infrastructures are designed more for end users to access data as needed, rather than for massively parallel systems to scan petabytes worth of data around the clock. To take advantage of Big Data, the infrastructure to access it has to scale in concurrent access speed and use intelligent storage and retrieval methods.

When scaling wide with lots of compute nodes generating lots of I/O throughput to access the various sources and locations that comprise Big Data repositories, you need to work smart.

The process of gathering raw data and manipulating it into useful repositories that can be accessed further down the line in Big Data analysis is called the *workflow*. Workflows can be either straightforward or very complicated. How complicated they become varies by what they are trying to accomplish, where the data comes from, and what kind of data is output, but all workflows require quick, consistent access to data. Many silos of data need to be sourced from

everywhere in your environment, and that means disparate systems all offering their own particular niche of information. And vice versa: multiple systems, each analyzing and generating specific views of all this data, need to eventually share all the information with the users. The workflow alternately storing and crunching Big Data is what brings these otherwise independent informational sources together to answer questions. Being able to easily access all the sources and take what you need quickly is key to the speed and success of Big Data. Avere Global Namespace simplifies data management and access in storage environments comprised of many storage systems. Avere is the only vendor to provide a global namespace that spans public object, private object, and legacy NAS from heterogeneous vendors.

Note: The entire end-to-end process involved in Big Data analysis depends on what industry you're in and how timely the information needs to be. Some Big Data workflows involve a huge amount of up-front data gathering from a multitude of sources. These "upstream" sources collect and analyze data individually and result in a simplified data set. Afterward, the processing "downstream" is just simple mathematics by compute nodes. Other workflows start as very simple pipelines of data from very few sources but accumulate vast amounts of data into a handful of extremely large repositories. Later, downstream in the workflow, many systems have to sift through these repositories simultaneously to find answers. Coordination of what data goes where and which system needs access to it at what time gets too complicated at the scale of Big Data. Data management tasks of Big Data get in the way of the real work that needs to be done.

Big Data analysis is about trying to predict the future or trying to find what went wrong. Sometimes Big Data is about simulating every possible option and picking the right one. Big Data is also about intelligently looking at risk, and avoiding it.

Whether animating the next Academy Award-winning film, hunting for oil, curing the deadliest diseases, or trying to pick the right investment strategy, Big Data needs to be accessed and processed to be worth the effort. Big Data analysis within the cloud using Avere Cloud NAS to access your data helps to solve these issues with an economical alternative to housing and managing the infrastructure yourself. This situation of vast amounts of data and vast amounts of compute power is perfect for cloud-based technology.

Utilizing Cloud Resources for Easier Data Management

Among the components of the cloud (refer to Chapter 2) is a compute farm. *Compute farms* are nothing more than a bunch of virtual server instances in the cloud that have a lot of CPU and memory configured on them for running complex software. This software can be custom written or off the shelf for working with your Big Data files. Using these virtual compute server instances, you can quickly spin up as many of them as you need and turn them on your Big Data files. When your analysis is done, you can spin down and delete the instances. You only pay for the time you use the compute farm. The entire virtual infrastructure exists only for the time you use it to churn through your Big Data files and come up with answers to your business questions.

Because Big Data is such a complicated set of operations in terms of analysis, it's difficult to apply one solution to every Big Data problem. The data access aspect of it is easily solved by using Avere Cloud NAS technology to intelligently manage the huge I/O burden involved in Big Data processing.

Chapter 5

Cloud NAS for Archiving

In This Chapter

- ▶ Looking at wasted space
- ▶ Demonstrating the cost effectiveness of cloud storage for data archival
- ▶ Making effortless data migration into and out of cloud storage
- ▶ Seeing how cloud-based archives radically reduce cost

This chapter covers the archival options that are available to you once you cloud-enable your NAS infrastructure with the Avere Cloud NAS solution. Archival options come in two general types:

- ✓ **Deep archive:** Most organizations associate this method with a long-term archive. I talk about using Amazon Glacier, a cloud-based archival solution, for this and tell you how it integrates with Avere Cloud NAS to quickly enable a long-term, or frozen, archival solution.
- ✓ **Active archive:** This method uses Avere FlashMove software to move data from a Core filer to cloud storage, where it can be more cost effective.

But, first things first: Before I talk about these two archive methods, you need to know why to archive data in the first place.

Most of Your Data Is Wasting Space

In general, everyone thinks that their data is important, pretty much because it's theirs and that's as good a reason as any. Data can be important at any given moment of

time, but because times change, so does the importance of data. A general principle is that only about 20 percent of a company's data is really timely. The other 80 percent just hangs around, wasting storage space. But don't take my word for it, this has been scientifically proven. This proof is called the *Pareto Principle*, more commonly known as the "80/20 rule," which applies to a wide range of economic and business situations.

One of the theorems of the Pareto Principle states that "80 percent of your business revenue comes from 20 percent of your customer base." Several theorems exist along these lines, but the one I use is this: In data processing environments, 80 percent of your users generally interact with only 20 percent of your data.



Although the 80/20 rule isn't a hard-and-fast rule, it strangely does prove itself out in most cases, especially when the environments grow significantly large and good practices around data management take second stage to getting business done. It's not that only one-fifth of your company's data has ever meant anything important to the company. It's that the sliding focus from new "hot" data to old "cold" data (refer to Chapter 3) over time creates a whole lot of unused data that you may want to keep but don't plan to use anytime soon. Knowing that 80 percent of your data is *much less critical* than the other 20 percent, you can use this information to your advantage. Tiering this less-used data to slower, and therefore cheaper, storage frees up space in faster storage for more active files.



I'm not saying that you should take the lion's share of your data and just upload it to the cloud. What I am saying is that a large chunk of your data probably isn't being stored where it should be, which would be the cloud. Moving cold data to a cloud-based tier makes sense because an Avere Cloud NAS cluster can hide the access latency for your clients to read those files from the cloud and the cost savings of cloud storage can be significant when compared to that of on-premise NAS filers. Furthermore, other files live within that 80 percent that you may never need to recall from the cloud, which is why a cloud-based archival solution such as Amazon Glacier may be the best place for it.

Cloud Storage Was Designed to be More Cost Effective

At the time of writing, Amazon charges \$0.01 per gigabyte (GB) per month for what you store in their long-term cloud storage solution, known as Glacier. That's right, just a penny per gigabyte per month! That's an incredibly low price to pay for data storage, and it's hard to find a more cost-effective place to store data long term.

Here's the catch.

First, Glacier is not typical read-write storage. It behaves more like a deep tape archive where you back up files to it and they sit collecting dust on a shelf somewhere. Because of the way Glacier operates, this is only for data that you expect to freeze for a *very* long time, potentially years or even decades. The intent is that, while you probably won't need those files ever again, you don't want to delete them either. Glacier is a long-term storage solution only, but it's the perfect complement to the performance scaling and optimization of Avere Cloud NAS, used as another tier of storage.

Second, while you don't have to pay to transmit data from your data center into the cloud, with Amazon cloud services, you *do* have to pay to retrieve data to put it back on your data center site. The cost to pull data from the cloud back into your enterprise can be significant. Before storing data in the cloud, you need to understand the data's usage pattern.

Data that is idle for long periods of time is best suited for cloud storage archiving. If your files are being read on a daily basis, however, you'll be getting charged for transferring data out of the cloud. The cost can add up, so you'd be smart to understand your usage before committing to the cloud. **Note:** The best indicator of when a file was opened last is the access time stamp.

Finally, when I talk about long-term data archival, I'm talking about data that will lie dormant for more than six months and, if recalled, doesn't have to reappear immediately or on demand; it just has to be available for access a couple of hours after it is requested. Amazon Glacier's return policy for recalled files is three to five hours, so it's definitely not real-time access — but keep in mind, you are only paying a penny per GB!

Moving Cold NAS Data to a Better Home in the Cloud

“Cold” data isn’t a de facto definition. *Cold* is a relative term and applies to files that have not been referenced in any way for a period of time — and most likely *never* will be accessed again. An analogy would be spring cleaning time around the house. You go around the house and find all the junk you’ve been accumulating during the year, or years in my case, and evaluate its worth to you. Although it isn’t as easy to judge the value of data as it is an old pair of ski boots, you do have some clear aspects of data files to use as guidelines, such as file-retention policies — which you definitely want to review prior to moving any files around. You also want to scrutinize other potential aspects of a file’s *worth* prior to actively migrating it to an archive.

Long-term deep archives

Deep archives operate under the expectation that your files will be very cold, as in not active, or as in the case of a glacier, frozen. Clever thinking, right? The folks at Amazon sure thought so when they came up with the name, Amazon Glacier!

Because Avere Cloud NAS treats the cloud like any other type of storage, moving data from legacy Core NAS filers to the cloud — and then into a frozen archive — is a very simple task.

As shown in Figure 5-1, Amazon Glacier is an extension of services from your already connected cloud services when using Amazon S3 for cloud storage.

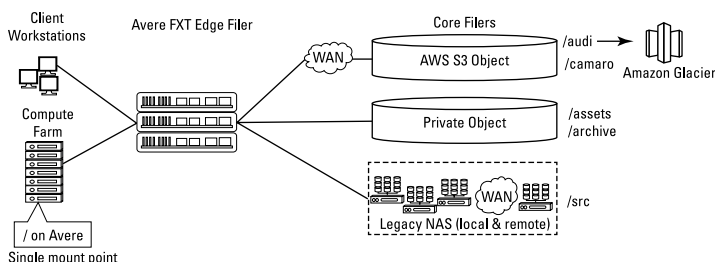


Figure 5-1: Where Amazon Glacier sits in the Avere Cloud NAS solution.

In the upper right of Figure 5-1, the “AWS S3 Object” would be your Amazon Web Services cloud storage. When you attach cloud storage to your Avere GNS, it appears like any Core filer storage folder. Amazon Glacier’s long-term storage is an optional tier of storage within their cloud. When you want to archive the data within a folder in the GNS, you can use the Avere admin GUI or command line interface (CLI) to “freeze” the folder.

Freezing a folder directs it to an Amazon Glacier archive tier of storage. The data in the folder appears to remain where it is, and you can do a directory listing and still see the filenames, size, and time stamps as usual. However, in order to read the files, an administrator has to go to the GUI or CLI and issue a thaw command to the folder and files you want to read or write to. The thaw command issues a restoration command to Amazon Glacier. Your files then take between three to five hours to return to a read/write state.

Freezing a folder in Avere moves the data from a standard cloud-storage bucket to a Glacier-enabled bucket in Amazon Web Services.



Glacier is meant to be for files that, in a paper-based office environment, would need to be boxed up and put on a shelf in a warehouse until someone needed them. To get those files back, you have to hire a courier to drive to the warehouse, find the right box, pull out the right file, and return it to the office. The process is very similar with Glacier, but at least it’s automated and secure.

Active archive

The active archive method is a data movement process that lets you move your files around to more cost-effective storage. With Avere Cloud NAS, cloud-storage buckets (refer to Chapter 2) act just like traditional NAS folders. All your storage, from your legacy NAS filers to cloud-based buckets, is mapped out under the Avere Global Namespace. This namespace presents your user community with a single view of your various storage silos, and moving data around doesn’t affect this view. Avere knows where your data resides at all times. This is a huge help to data-management tasks, because one of the biggest headaches of data placement is being able to find everything after you move it. Knowing that you can

migrate data from one type of storage to another, for example from your expensive Core NAS filer or lower-cost white-box NAS filer to an even lower-cost cloud-storage bucket, without remapping your users to the new spot is a huge benefit.

Figure 5-2 shows an example of how archive plays into this. You can see how Avere FlashMove software moves the “/archive” folder from an in-house NAS filer to cloud storage.

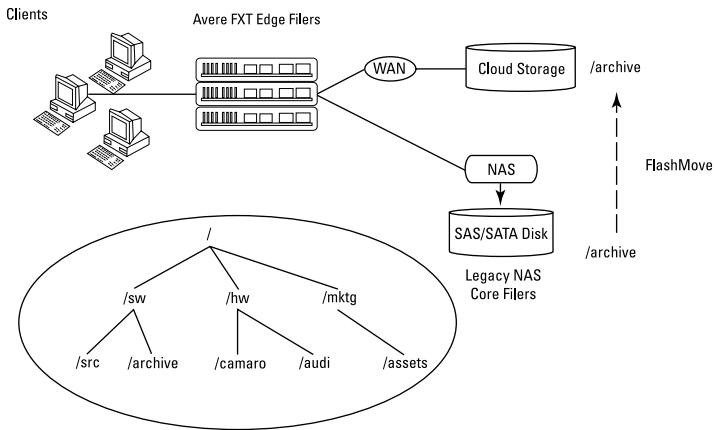


Figure 5-2: Use FlashMove feature to archive to cost-saving cloud storage.

In Figure 5-2, the full path name that users recognize to reference the “/archive” folder of “/sw/archive” is left unchanged. The user never knows that anything has moved. The actual files in that “/archive” folder will migrate behind the scenes from a NAS filer to cloud storage (or back again if required). All access to and from those files will occur from the Avere Edge filers to the new location in the cloud. The added benefit is that you have more space on the NAS Core filer released by the files migrated from “/archive” that now occupy cloud storage.

Using a Cloud NAS solution gives you a few options of how to store data long term, as well as how to transparently migrate data to lower tiers to free up faster storage resources.

TCO Studies Show the Hybrid Cloud Savings as an Archive

When a company is looking to save money, a few pennies here and there aren't worth pursuing. You want to show big savings, preferably immediately, as well as ongoing savings over a short period of time, say three to five years.

Archiving data using one of the two methods I describe earlier immediately results in savings. Once files are moved from the Core filer to the cloud, the space on the Core filer is freed up for new files. Because of the price difference between your legacy NAS Core filer storage and less-costly cloud storage, you can delay the need to buy more storage to expand the Core filers. Those delayed or avoided purchases of more expensive Core filer storage can add up to significant savings.

Total Cost of Ownership (TCO) evaluates how much you're paying for a particular solution. When you perform a TCO study, the hardware, software, services, and support are all factored into the analysis, along with the cost of operating the solution over time. This gives you a nice chart of numbers to compare and contrast different solution options.

When comparing the pricing of various solutions, the dollar-per-gigabyte of storage is not the only factor. In fact, the dramatically lower cost of storage that cloud providers charge highlights other hidden costs that need focusing on to make an informed decision. These hidden costs are things that are often taken for granted. Power, cooling, and even the physical space to house your storage environment cost money, and these costs can add up to either savings or expenses if you aren't careful to factor them in to the TCO study.

Here's a list of cost factors that contribute to the TCO of cloud-based archive solutions and what you need to evaluate in each area:

- ✓ **Storage to house archive data:** Cloud storage costs substantially less with this since you have no capital equipment expense but rather a monthly fee for the amount of capacity stored in the cloud.

- ✓ **Equipment maintenance:** These costs qualify as hidden costs because you don't realize they're there until it's time to renew the maintenance contract for aging equipment, which is sure to come up in a long-lived archive environment.
- ✓ **Administration:** These costs apply to the people monitoring, managing, and operating the systems. People are an expensive resource.
- ✓ **Charges for uploading and downloading data in the cloud:** This cost is based on usage so you need to estimate activity expectations.
- ✓ **Facilities charges (power, cooling, and space):** Facilities charges are a big cost savings when going with the cloud, because you're not paying directly for any equipment.

Using the preceding list of factors is critical to understanding how a cloud-based archive would benefit your organization. The costs of storage, facilities, and administration typically have been the deciding factor in most TCO assessments. For cloud-based archiving, with Avere Cloud NAS, you can reduce those costs drastically. You also can demonstrate the cost savings of introducing cloud storage into your environment, which allows you the flexibility and confidence to archive data efficiently and cost effectively.

Chapter 6

Ten Benefits of Cloud NAS

In This Chapter

- Checking out the good things about Cloud NAS

This chapter highlights how a Cloud NAS solution specifically helps to tackle both the daily problems with growing and managing storage and long-term issues of trying to keep up with explosive data growth and costs in your organization.

- ✓ **Getting performance where you need it:** With Avere FXT Edge filers tracking the usage patterns of your files and moving the most active files closer to the clients using them, you decrease latency to clients and reduce the burden on your Core NAS filers, optimizing performance.
- ✓ **Keeping pace with growing demand:** Demand for more application performance is growing every year, and this places an increasing demand on your storage. Avere FXT Edge filers support clustering from 3 to 50 nodes to help you keep pace.
- ✓ **Storing data where it works best and costs the least:** Because the Avere Cloud NAS Edge-Core architecture is aware of multiple types of storage media, each with its own performance and cost characteristics, your data will always be placed where it makes the most sense, technologically *and* economically — whether that's on premise or in the cloud.
- ✓ **Making sure that your data is always available:** Avere FXT clusters support high availability (HA) software that ensures that your data is always available even in the presence of network, hardware, and other failures.
- ✓ **Managing your data as a single pool:** Storage silos be gone! Avere Global Namespace integrates all your Core filers, both NAS and cloud, into a single pool of storage with transparent migration between them.

- ✓ **Keeping your data safe:** Replicate your data for protection from unexpected disasters, such as floods, fire, or regional infrastructure issues (power outages from storms, for example). Mirroring software built into the Avere Edge filers can keep a safe copy up to date anywhere in the global namespace it manages.
- ✓ **Enjoying global collaboration:** By being able to put data in the cloud, your data becomes cloud accessible and can be used as a shared access point. Other remote offices, or even other companies, can collaborate in data-processing tasks without the trouble of physically joining their corporate networks with yours.
- ✓ **Protecting your investment:** Cloud NAS augments and extends the capabilities of your existing NAS environments so that you're not replacing any gear you've already purchased or are still paying for.
- ✓ **Consolidating and centralizing management:** By gathering together existing NAS environments under the umbrella of Avere Edge filers, you get a bird's eye consolidated view of the performance of your core NAS filers (and awesome reporting), all from one place instead of several individual management consoles.
- ✓ **Understanding usage and data flow better:** The ability to view what data is being accessed and with what regularity greatly increases your understanding of what your active and inactive storage areas are. This information helps you plan future purchases with more insight into where to focus your storage spending.

Get your NAS storage connected to the cloud!

Today, organizations have huge investments in their existing NAS storage infrastructures. The increasing pressure to do more with less and the rise of cloud storage as an alternative to traditional NAS is forcing many IT departments to reinvent the wheel. This book explores another approach — cloud on demand — by offering a combined solution of cloud storage and NAS called Cloud NAS.

- **Protect your investment** — *keep your Core NAS filers online and offload processing and data to the Edge*
- **Redistribute your data** — *move your data to where it works smarter on your current infrastructure without migration downtime*
- **Jump to the cloud** — *make private and public cloud solutions work for you without reinventing the wheel*
- **Change without disruption** — *enable scalable performance and cloud accessibility to your applications without rewriting code and retraining users and staff*

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Open the book and find:

- How to optimize your existing NAS environments
- An explanation of cloud storage concepts and benefits
- How to easily connect your storage silos to the cloud
- Ways to build Big Data and cloud-based archiving solutions

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